

HKJC Race-Day Projection Model — v4.4

Technical Overview & Acronym Reference

1. Model Overview

The model projects finish times for every runner in an upcoming HKJC meeting. Each horse's projection combines a **baseline expected time** (ET) for its race conditions with a **personalised residual** derived from its historical performance, plus adjustments for draw, sectional profile, and predicted race position. Horses are ranked by projected time; win probabilities are computed via a Boltzmann distribution.

The pipeline runs five sequential stages: (1) parse the race card, (2) load ET reference tables and compute draw offsets from the historical database, (3) precompute sectional zone deviations for all horses, (4) project each race, (5) generate outputs (PDF, text, JSON).

2. Projection Pipeline

2.1 Expected Time (ET) Lookup

ET is the median historical finish time for a given distance, going, weight band, course configuration, and class (or track type). It acts as the **baseline** — what an average horse would run under today's conditions. The 4-tier cascade tries progressively coarser groupings until it finds a reference with sufficient sample size:

```
ClassFine (n≥2): distance + going + weight band + course + class
Fine (n≥5): distance + going + weight band + course (all classes pooled, with class-band correction offset)
Coarse (n≥5): distance + going + weight band + track type
Ultra (n≥5): distance + going + track type (weight band dropped)
```

Weight is discretised into 5 bands: 105-115, 116-120, 121-125, 126-130, 131-135 lbs (after subtracting apprentice claims).

2.2 Recency Residual (Ability Signal)

For each of the horse's historical runs, the model computes a **contextualised residual**:

```
residual = finish_time - ET(run conditions) - race_pace_index + position_credit -
draw_offset - wide_running_cost
```

This removes race-level pace effects (tactical/slow races), rewards good finishing positions, and strips out trip luck (draw savings or wide-running costs). The residuals capture the horse's **true latent ability**.

Residuals are then combined into a single number via **recency-weighted averaging** ($\lambda = 0.85$ exponential decay — most recent run gets full weight, each prior run discounted 15%). Additional discounts apply for cross-surface (AWT→Turf ×0.50), cross-venue (HV↔ST ×0.50), and asymmetric distance penalties (stepping up penalised more than down). The most recent win gets a ×2 boost.

2.3 Projection Formula

```
projection = ET(today's conditions) + recency_residual + draw_offset + uncertainty_penalty
+ surface_penalty + class_transition_adj + sectional_adj + speed_map_positional_adj
```

No shrinkage toward the class mean is applied — the horse's own data drives the entire projection (**ability-anchored**). Thin-data uncertainty is handled by additive penalties, not by diluting the signal.

2.4 Draw Offset

For each (distance, course, draw) combination, the historical database provides a raw offset computed via the **race-median method**: $\text{residual} = \text{finish_time} - \text{race_median_time}$, then the draw's mean residual minus the overall

mean. This normalises within each race, removing confounders like horse quality.

The raw offset is Bayesian-shrunk toward zero: $\text{adj} = (n / (n + 8)) \times \text{raw}$. Outer draws in large fields (≥ 12 runners) are amplified by a field-size factor. Safety cap: $\pm 1.50\text{s}$.

2.5 Sectional Decomposition (v4.4)

Each historical run's sectional times are split into Early / Mid / Late zones (per-400m normalised), and deviations from the race median are computed. A horse's **sectional profile** (last 6 runs, recency-weighted) captures its finishing pattern.

Three components adjust the projection:

- (a) **Late finishing signal**: $\text{avg_late_dev} \times 0.12$ (strong finishers save time)
- (b) **Distance suitability**: front-loaded SSI > 0.15 at $\geq 1600\text{m}$ penalised $\times 0.10$ (fade risk); sprints: $\text{SSI} \times 0.04$
- (c) **Reliability**: late_std above 0.30 adds $+0.02/\text{unit}$ (inconsistent finishers)

2.6 Speed Map

A 3-row $\times$ N-column grid predicts early-race positions. Columns (front $\rightarrow$ back) sorted by ESZ; rows (rail $\rightarrow$ wide) sorted by draw within each column. Positional advantage scored by style-position alignment, rail-saving at bends, and wide draw costs.

Two adjustment components:

- (a) **Positional cost**: $\text{advantage} \times 0.06\text{s}$ (capped $\pm 0.15\text{s}$)
- (b) **Pace $\times$ Position $\times$ Style (PPS)**: 36-entry lookup table mapping (pace bucket, grid row, running style) to seconds

2.7 Win Probability

Boltzmann distribution: $p(i) \propto \exp(-(\text{time}_i - \text{time}_{\min}) / T)$, where $T = 0.80 \times \sigma(\text{field times})$. Class-tier scaling widens T for C4 ($\times 1.30$) and C5 ($\times 1.50$) where model precision is lower. Low-SF horses get confidence-deflated probabilities.

3. Penalty & Adjustment System

Penalty / Adjustment	Trigger	Effect
Uncertainty (U+)	$n_runs < 4$	$+0.10/\sqrt{n}$ (strong debut: $+0.06/\sqrt{n}$)
Surface mismatch (SP+)	Few/no same-surface runs	$+0.18\text{s}$ (0 runs) or $+0.12/\sqrt{n}$
Class step-up (CTP+)	Modal class $<$ today's class	Penalty scales with ET gap – old-class margin
Class step-down	Modal class $>$ today's class	Bonus: $-\text{ET_gap} \times 0.60 \times \text{dominance}$
Sectional late adj	≥ 2 profiled runs	$\pm \text{avg_late_dev} \times 0.12\text{s}$
Sectional SSI dist	$\text{SSI} > 0.15$ at $\geq 1600\text{m}$	$+(\text{SSI} - 0.15) \times 0.10\text{s}$
Speed map position	All projected horses	$-\text{advantage} \times 0.06\text{s}$ ($\pm 0.15\text{s}$ cap)
Pace $\times$ Position $\times$ Style	All projected horses	Lookup: -0.06s to $+0.05\text{s}$

4. Trajectory Detection (Informational)

Horses are classified as Improving / Declining / Stable / Insufficient based on last-2 runs vs prior-3 baseline + OLS slope confirmation. Trajectory is **reported in commentary only** — no time adjustment is applied (removed v4.1, as recency $\lambda=0.85$ already captures form trends).

5. Acronym & Column Reference

Term	Meaning
ET	Expected Time — baseline finish time for conditions (from 4-tier cascade)
Proj T	Projected Time — ET + residual + all adjustments. Lower = faster = better.
Win%	Win probability from Boltzmann distribution. Confidence-deflated for thin data.
Ability	Mean raw residual (finish_time – ET) across all runs. Negative = faster than ET.
Eff Resid	Effective Residual — recency-weighted, contextualised. The key ability signal.
SF	Shrinkage Factor = $n/(n+5)$. Higher = more data = more trusted. 0.50+ = high confidence.
Draw Off	Bayesian-shrunk draw offset in seconds. Positive = disadvantageous draw.
SSI	Sectional Speed Index = late_dev – early_dev. Negative = strong finisher. Positive = front-loaded (fade risk at distance).
Fin Sec	Projected final 400m sectional time. Source indicators: (blank)=horse, ~=blend, c=class.
ESZ	Early Speed Z-score. Negative = naturally fast early. Used for speed map placement.
Style	Dominant running style: Leader / On-Pace / Midfield / Closer / Unknown.
Vet	Veterinary flag from vet report: RED / AMB(er) / INF(o) / blank.
U+	Uncertainty penalty applied ($n < 4$ runs).
SP+	Surface mismatch penalty (few same-surface runs).
CTP+	Class Transition Penalty (stepping up in class).
↑IMP	Improving trajectory detected (informational).
↓DEC	Declining trajectory detected (informational).
PPS	Pace × Position × Style interaction adjustment from speed map.
RPI	Race Pace Index — median field residual for a historical race. Subtracted from each run to remove tactical pace effects.
λ	Recency decay parameter (0.85). Each prior run's weight = $\lambda^{(n-1-i)}$.
DB	Historical database (~18,000 race-runs from HKJC).

6. Key Tunable Constants

Constant	Value	Purpose
RECENCY_LAMBDA	0.85	Exponential decay per prior run
UNCERTAINTY_BASE	0.10s	Thin-data penalty coefficient
DRAW_SHRINKAGE_K	8	Bayesian shrinkage strength for draws
SEC_LATE_DEV_COEFF	0.12	Late finishing signal weight
SEC_SSI_DIST_COEFF	0.10	SSI distance-suitability weight ($\geq 1600\text{m}$)
SEC_SSI_SPRINT_COEFF	0.04	SSI sprint weight ($\leq 1200\text{m}$)
SMAP_TIME_COEFF	0.06	Speed map advantage → seconds conversion
SMAP_TIME_CAP	0.15s	Max positional adjustment
AWT_TURF_DISCOUNT	0.50	AWT run weight when today is Turf
HV_ST_DISCOUNT	0.50	Cross-venue run weight discount
LAST_WIN_BOOST	2.0	Recency weight multiplier for last-start winner
POSITION_CREDIT	-0.15 / -0.08 / -0.04	Credit for 1st / 2nd / 3rd place

7. PDF Report Column Layout

Each race page contains an 18-column ranking table (Rk, No, Horse, Wt, Dr, Rtg, ExpT, ProjT, Win%, Ability, EffResid, SF, DrawOff, SSI, FinSec, ESZ, Style, Vet), followed by a speed map grid (3 rows × N columns, green = advantage, red = disadvantage), positional notes, up to 4 beneficiaries, and commentary bullets.

Top-3 rows are shaded green/blue/yellow. Win% ≥20% is bolded green.

8. Module Interaction Diagram

```
Race Card → horse_no, weight, draw, jockey  
ET Refs → expected_time for today's conditions  
Historical DB → all past runs → recency_residual, draw_offset, horse_profile (style, SF, ESZ), sectional_deviations  
project_race() combines: ET + recency_resid + draw_off + penalties + sectional_adj + smap_adj → projected_time  
enrich_with_risk() → win_prob (Boltzmann)  
compute_speed_map() → grid placement → smap_time_adjustment  
race_commentary() → interpretive bullets  
generate_pdf() / generate_text_report() → outputs
```